# Supplementary material

All analyses use corrected selection unless labelled historical. Rates and slope units are deaths per 100,000; APC is percent per year. Full-precision CSV results and model warnings are delivered with the code.

## Table S1. Corrected selection by year

| Year | All SIM | J45/J46 | COVID | Duplicates | Invalid dates | Negative age | Valid | ≥7 | 5–34 |
|---|---|---|---|---|---|---|---|---|---|
| 2014 | 1227039 | 2129 | 0 | 0 | 2 | 0 | 2127 | 2035 | 182 |
| 2015 | 1264175 | 2232 | 0 | 0 | 1 | 0 | 2231 | 2142 | 189 |
| 2016 | 1309774 | 2249 | 0 | 0 | 2 | 0 | 2247 | 2187 | 184 |
| 2017 | 1312663 | 2477 | 0 | 0 | 1 | 0 | 2476 | 2418 | 172 |
| 2018 | 1316719 | 2307 | 0 | 0 | 4 | 0 | 2303 | 2244 | 200 |
| 2019 | 1349801 | 2488 | 0 | 0 | 2 | 0 | 2486 | 2419 | 218 |
| 2020 | 1556824 | 2735 | 27 | 0 | 3 | 0 | 2705 | 2677 | 236 |
| 2021 | 1832649 | 2488 | 30 | 0 | 0 | 0 | 2458 | 2422 | 209 |

COVID = union across five cause lines. Invalid-date and negative-age counts follow COVID and deduplication. Both ambiguous 2018 records are retained. The historical selection, provided separately, removed 56 COVID records, one record under the reduced key and 15 records with invalid dates, leaving 18,583 deaths at continuous age >6.

## Table S2. Annual Aligned crude ≥7

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2014 | 2035 | 181,252,457 | 1.122743 |
| 2015 | 2142 | 182,968,415 | 1.170694 |
| 2016 | 2187 | 184,647,932 | 1.184416 |
| 2017 | 2418 | 186,273,291 | 1.298093 |
| 2018 | 2244 | 187,895,171 | 1.194283 |
| 2019 | 2419 | 189,505,848 | 1.276478 |
| 2020 | 2677 | 191,083,309 | 1.400960 |
| 2021 | 2422 | 192,673,672 | 1.257048 |

## Table S2. Annual Brazil 2014 ASR ≥7

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2014 | 2035 | 181,252,457 | 1.122743 |
| 2015 | 2142 | 182,968,415 | 1.144431 |
| 2016 | 2187 | 184,647,932 | 1.132194 |
| 2017 | 2418 | 186,273,291 | 1.208506 |
| 2018 | 2244 | 187,895,171 | 1.089633 |
| 2019 | 2419 | 189,505,848 | 1.140337 |
| 2020 | 2677 | 191,083,309 | 1.229760 |
| 2021 | 2422 | 192,673,672 | 1.077177 |

## Table S2. Annual Crude 5–34

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2014 | 182 | 100,130,699 | 0.181762 |
| 2015 | 189 | 99,610,503 | 0.189739 |
| 2016 | 184 | 99,048,598 | 0.185767 |
| 2017 | 172 | 98,452,312 | 0.174704 |
| 2018 | 200 | 97,819,731 | 0.204458 |
| 2019 | 218 | 97,214,347 | 0.224247 |
| 2020 | 236 | 96,671,200 | 0.244126 |
| 2021 | 209 | 96,070,603 | 0.217548 |

## Table S2. Annual Brazil 2014 ASR 5–34

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2014 | 182 | 100,130,699 | 0.181762 |
| 2015 | 189 | 99,610,503 | 0.189060 |
| 2016 | 184 | 99,048,598 | 0.184945 |
| 2017 | 172 | 98,452,312 | 0.174255 |
| 2018 | 200 | 97,819,731 | 0.203444 |
| 2019 | 218 | 97,214,347 | 0.222528 |
| 2020 | 236 | 96,671,200 | 0.241039 |
| 2021 | 209 | 96,070,603 | 0.214535 |

## Table S2. Annual Crude 7–34

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2014 | 171 | 94,235,685 | 0.181460 |
| 2015 | 174 | 93,780,394 | 0.185540 |
| 2016 | 179 | 93,255,055 | 0.191947 |
| 2017 | 164 | 92,641,638 | 0.177026 |
| 2018 | 190 | 92,007,559 | 0.206505 |
| 2019 | 208 | 91,371,402 | 0.227642 |
| 2020 | 232 | 90,729,128 | 0.255706 |
| 2021 | 201 | 90,129,917 | 0.223011 |

## Table S2. Annual Crude 35–59

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2014 | 452 | 62,893,940 | 0.718670 |
| 2015 | 485 | 64,149,669 | 0.756044 |
| 2016 | 515 | 65,398,428 | 0.787481 |
| 2017 | 567 | 66,640,781 | 0.850830 |

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2018 | 504 | 67,862,310 | 0.742680 |
| 2019 | 524 | 69,039,371 | 0.758987 |
| 2020 | 709 | 70,157,104 | 1.010589 |
| 2021 | 591 | 71,213,520 | 0.829899 |

## Table S2. Annual Crude ≥60

| Year | Deaths | Population | Rate |
|---|---|---|---|
| 2014 | 1412 | 24,122,832 | 5.853376 |
| 2015 | 1483 | 25,038,352 | 5.922914 |
| 2016 | 1493 | 25,994,449 | 5.743534 |
| 2017 | 1687 | 26,990,872 | 6.250261 |
| 2018 | 1550 | 28,025,302 | 5.530716 |
| 2019 | 1687 | 29,095,075 | 5.798232 |
| 2020 | 1736 | 30,197,077 | 5.748901 |
| 2021 | 1630 | 31,330,235 | 5.202642 |

## Table S3. Historical age-group reproduction

| series | n_2014 | n_2021 | rate_2014 | rate_2021 | pct_change_2014_2021 | slope | p_value |
|---|---|---|---|---|---|---|---|
| legacy_lt18 | 43 | 37 | 0.07649 | 0.06945 | -9.19335 | -0.00047 | 0.85236 |
| legacy_18_59 | 580 | 756 | 0.47786 | 0.58735 | 22.91210 | 0.02125 | 0.04388 |
| legacy_gte60 | 1412 | 1630 | 5.85338 | 5.20264 | -11.11724 | -0.07124 | 0.13500 |

Historical <18 numerator is floor ages 7–17 and denominator ages 0–17. These rows reproduce the historical construction and are not corrected primary domains.

## Table S4. Count models, IBGE 2018

| Series | Family | APC | 95% CI | p | Dispersion | Theta | Status |
|---|---|---|---|---|---|---|---|
| Crude ≥7 | Poisson | 2.190 | 1.549, 2.835 | <0.001 | 6.04 | | ok |
| Crude ≥7 | Quasi-Poisson | 2.190 | 0.236, 4.183 | 0.033 | 6.04 | | ok |
| Crude ≥7 | Negative binomial | 2.221 | 0.886, 3.574 | 0.001 | 1.33 | 686.8 | ok |
| Crude 5–34 | Poisson | 3.998 | 1.787, 6.258 | <0.001 | 1.09 | | ok |
| Crude 5–34 | Quasi-Poisson | 3.998 | 1.130, 6.949 | 0.014 | 1.09 | | ok |
| Crude 5–34 | Negative binomial | 3.998 | 1.787, 6.258 | <0.001 | 1.09 | 3280138.3 | warning |
| Crude 7–34 | Poisson | 4.604 | 2.327, 6.931 | <0.001 | 1.15 | | ok |
| Crude 7–34 | Quasi-Poisson | 4.604 | 1.561, 7.738 | 0.010 | 1.15 | | ok |
| Crude 7–34 | Negative binomial | 4.604 | 2.327, 6.931 | <0.001 | 1.15 | 2427388.1 | warning |
| Crude 35–59 | Poisson | 2.779 | 1.448, 4.126 | <0.001 | 5.54 | | ok |
| Crude 35–59 | Quasi-Poisson | 2.779 | -1.082, 6.789 | 0.130 | 5.54 | | ok |
| Crude 35–59 | Negative binomial | 2.783 | 0.181, 5.453 | 0.036 | 1.34 | 187.9 | ok |

| Series | Family | APC | 95% CI | p | Dispersion | Theta | Status |
|---|---|---|---|---|---|---|---|
| Crude ≥60 | Poisson | -1.300 | -2.047, -0.546 | <0.001 | 3.55 | | ok |
| Crude ≥60 | Quasi-Poisson | -1.300 | -3.049, 0.482 | 0.124 | 3.55 | | ok |
| Crude ≥60 | Negative binomial | -1.271 | -2.480, -0.046 | 0.042 | 1.34 | 972.6 | ok |
| Age-adjusted ≥7 | Poisson | 0.001 | -0.626, 0.632 | 0.998 | 3.11 | | ok |
| Age-adjusted ≥7 | Quasi-Poisson | 0.001 | -1.125, 1.140 | 0.999 | 3.11 | | ok |
| Age-adjusted ≥7 | Negative binomial | 0.926 | -0.197, 2.061 | 0.106 | 1.30 | 147.8 | ok |

Rows marked as warnings reached the theta iteration limit; convergence of mean-model IRLS does not prove convergence of theta. Retain these estimates with caution. The recorded warning was “iteration limit reached”. All errors, warnings, iterations, coefficients and fitted objects are retained in CSV/RDS. Quasi-Poisson uses t intervals with residual degrees of freedom; Poisson and negative binomial use normal Wald intervals.

# Table S4. Count models, IBGE 2024

| Series | Family | APC | 95% CI | p | Dispersion | Theta | Status |
|---|---|---|---|---|---|---|---|
| Crude ≥7 | Poisson | 2.312 | 1.670, 2.958 | <0.001 | 5.98 | | ok |
| Crude ≥7 | Quasi-Poisson | 2.312 | 0.366, 4.297 | 0.027 | 5.98 | | ok |
| Crude ≥7 | Negative binomial | 2.342 | 1.012, 3.690 | <0.001 | 1.33 | 695.6 | ok |
| Crude 5–34 | Poisson | 4.178 | 1.964, 6.441 | <0.001 | 1.09 | | ok |
| Crude 5–34 | Quasi-Poisson | 4.178 | 1.303, 7.135 | 0.012 | 1.09 | | ok |
| Crude 5–34 | Negative binomial | 4.178 | 1.964, 6.441 | <0.001 | 1.09 | 3258238.9 | warning |
| Crude 7–34 | Poisson | 4.791 | 2.511, 7.122 | <0.001 | 1.16 | | ok |
| Crude 7–34 | Quasi-Poisson | 4.791 | 1.742, 7.931 | 0.008 | 1.16 | | ok |
| Crude 7–34 | Negative binomial | 4.791 | 2.511, 7.122 | <0.001 | 1.16 | 2402052.5 | warning |
| Crude 35–59 | Poisson | 2.843 | 1.511, 4.192 | <0.001 | 5.52 | | ok |
| Crude 35–59 | Quasi-Poisson | 2.843 | -1.013, 6.849 | 0.123 | 5.52 | | ok |
| Crude 35–59 | Negative binomial | 2.844 | 0.246, 5.510 | 0.032 | 1.34 | 189.0 | ok |
| Crude ≥60 | Poisson | -1.210 | -1.959, -0.455 | 0.002 | 3.33 | | ok |
| Crude ≥60 | Quasi-Poisson | -1.210 | -2.907, 0.517 | 0.136 | 3.33 | | ok |
| Crude ≥60 | Negative binomial | -1.186 | -2.359, 0.001 | 0.050 | 1.34 | 1080.7 | ok |
| Age-adjusted ≥7 | Poisson | 0.290 | -0.340, 0.924 | 0.368 | 3.02 | | ok |
| Age-adjusted ≥7 | Quasi-Poisson | 0.290 | -0.823, 1.415 | 0.606 | 3.02 | | ok |
| Age-adjusted ≥7 | Negative binomial | 1.137 | 0.031, 2.256 | 0.044 | 1.31 | 156.0 | ok |

Rows marked as warnings reached the theta iteration limit; convergence of mean-model IRLS does not prove convergence of theta. Retain these estimates with caution. The recorded warning was “iteration limit reached”. All errors, warnings, iterations, coefficients and fitted objects are retained in CSV/RDS. Quasi-Poisson uses t intervals with residual degrees of freedom; Poisson and negative binomial use normal Wald intervals.

# Table S5. Standards and domains, IBGE 2018

| Series | 2014 | 2021 | Δ rate | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Brazil ASR 35–59 | 0.71867 | 0.82485 | +0.10618 | +14.77 | 0.02134 (-0.00957, 0.05225) | 0.142 |
| Brazil ASR 5–34 | 0.18176 | 0.21453 | +0.03277 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018 |

| Series | 2014 | 2021 | Δ rate | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Brazil ASR 7–34 | 0.18146 | 0.21955 | +0.03809 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013 |
| Brazil ASR ≥60 | 5.85338 | 5.08539 | -0.76799 | -13.12 | -0.08864 (-0.18714, 0.00986) | 0.070 |
| Brazil ASR ≥7 | 1.12274 | 1.07718 | -0.04557 | -4.06 | 0.00016 (-0.02139, 0.02170) | 0.986 |
| WHO ASR 35–59 | 0.71921 | 0.82668 | +0.10747 | +14.94 | 0.02162 (-0.00936, 0.05260) | 0.139 |
| WHO ASR 5–34 | 0.17788 | 0.20702 | +0.02914 | +16.38 | 0.00678 (0.00140, 0.01216) | 0.022 |
| WHO ASR 7–34 | 0.17730 | 0.21246 | +0.03516 | +19.83 | 0.00819 (0.00235, 0.01403) | 0.014 |
| WHO ASR ≥60 | 5.55943 | 4.81550 | -0.74393 | -13.38 | -0.08637 (-0.17725, 0.00451) | 0.059 |
| WHO ASR ≥7 | 1.09451 | 1.04795 | -0.04656 | -4.25 | -0.00015 (-0.02077, 0.02047) | 0.986 |

# Table S5. Standards and domains, IBGE 2024

| Series | 2014 | 2021 | Δ rate | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Brazil ASR 35–59 | 0.72112 | 0.83140 | +0.11028 | +15.29 | 0.02192 (-0.00907, 0.05291) | 0.134 |
| Brazil ASR 5–34 | 0.18281 | 0.21972 | +0.03691 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014 |
| Brazil ASR 7–34 | 0.18258 | 0.22505 | +0.04247 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011 |
| Brazil ASR ≥60 | 5.88365 | 5.30051 | -0.58314 | -9.91 | -0.06471 (-0.16221, 0.03278) | 0.155 |
| Brazil ASR ≥7 | 1.12844 | 1.11123 | -0.01722 | -1.53 | 0.00393 (-0.01762, 0.02547) | 0.671 |
| WHO ASR 35–59 | 0.72095 | 0.83282 | +0.11187 | +15.52 | 0.02223 (-0.00879, 0.05326) | 0.130 |
| WHO ASR 5–34 | 0.17884 | 0.21201 | +0.03317 | +18.55 | 0.00744 (0.00193, 0.01294) | 0.016 |
| WHO ASR 7–34 | 0.17835 | 0.21778 | +0.03943 | +22.11 | 0.00889 (0.00290, 0.01487) | 0.011 |
| WHO ASR ≥60 | 5.60811 | 5.01048 | -0.59763 | -10.66 | -0.06712 (-0.15761, 0.02337) | 0.119 |
| WHO ASR ≥7 | 1.10228 | 1.07940 | -0.02288 | -2.08 | 0.00305 (-0.01761, 0.02370) | 0.731 |

# Table S6. Kitagawa, IBGE 2018

| Endpoints | Crude start | Crude end | Net Δ | Composition | Rate term | Composition % |
|---|---|---|---|---|---|---|
| 2014–2021 | 1.122743 | 1.257048 | +0.134304 | +0.191434 | -0.057129 | 142.54 |
| 2014–2019 | 1.122743 | 1.276478 | +0.153734 | +0.138122 | +0.015612 | 89.84 |
| 2014–2020 | 1.122743 | 1.400960 | +0.278216 | +0.171157 | +0.107059 | 61.52 |
| 2015–2021 | 1.170694 | 1.257048 | +0.086354 | +0.167391 | -0.081037 | 193.84 |
| 2016–2021 | 1.184416 | 1.257048 | +0.072632 | +0.139935 | -0.067304 | 192.66 |
| 2014–2018 | 1.122743 | 1.194283 | +0.071540 | +0.107366 | -0.035827 | 150.08 |

Each term is independently rounded. The unrounded identity is tested to 1e−12; rounded terms may differ in their last displayed digit. A share over 100% reflects a negative rate term, not a probability.

# Table S6. Kitagawa, IBGE 2024

| Endpoints | Crude start | Crude end | Net Δ | Composition | Rate term | Composition % |
|---|---|---|---|---|---|---|
| 2014–2021 | 1.128443 | 1.274118 | +0.145675 | +0.170443 | -0.024768 | 117.00 |
| 2014–2019 | 1.128443 | 1.290082 | +0.161639 | +0.130868 | +0.030771 | 80.96 |
| 2014–2020 | 1.128443 | 1.417122 | +0.288679 | +0.158796 | +0.129883 | 55.01 |
| 2015–2021 | 1.177572 | 1.274118 | +0.096546 | +0.148180 | -0.051634 | 153.48 |
| 2016–2021 | 1.192574 | 1.274118 | +0.081544 | +0.121796 | -0.040252 | 149.36 |
| 2014–2018 | 1.128443 | 1.205894 | +0.077451 | +0.101345 | -0.023893 | 130.85 |

Each term is independently rounded. The unrounded identity is tested to 1e−12; rounded terms may differ in their last displayed digit. A share over 100% reflects a negative rate term, not a probability.

## Table S7. Pandemic-period influence, age_5_34, IBGE 2018

| Scenario | Series | Years | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|
| Principal | Brazil ASR 5–34 | 8 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018 |
| Principal | Crude 5–34 | 8 | +19.69 | 0.00795 (0.00212, 0.01377) | 0.016 |
| Principal | WHO ASR 5–34 | 8 | +16.38 | 0.00678 (0.00140, 0.01216) | 0.022 |
| Without 2020 | Brazil ASR 5–34 | 7 | +18.03 | 0.00581 (0.00040, 0.01122) | 0.040 |
| Without 2020 | Crude 5–34 | 7 | +19.69 | 0.00619 (0.00070, 0.01167) | 0.034 |
| Without 2020 | WHO ASR 5–34 | 7 | +16.38 | 0.00536 (-0.00017, 0.01089) | 0.055 |
| 2014–2019 | Brazil ASR 5–34 | 6 | +22.43 | 0.00675 (-0.00228, 0.01578) | 0.107 |
| 2014–2019 | Crude 5–34 | 6 | +23.37 | 0.00701 (-0.00219, 0.01622) | 0.102 |
| 2014–2019 | WHO ASR 5–34 | 6 | +22.80 | 0.00660 (-0.00250, 0.01570) | 0.114 |

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

## Table S7 continued. Count models, age_5_34, IBGE 2018

| Scenario | Family | APC | 95% CI | p | df | Status |
|---|---|---|---|---|---|---|
| Principal | Poisson | 3.998 | 1.787, 6.258 | <0.001 | 6 | ok |
| Principal | Quasi-Poisson | 3.998 | 1.130, 6.949 | 0.014 | 6 | ok |
| Principal | Negative binomial | 3.998 | 1.787, 6.258 | <0.001 | 6 | warning |
| Without 2020 | Poisson | 3.159 | 0.743, 5.633 | 0.010 | 5 | ok |
| Without 2020 | Quasi-Poisson | 3.159 | 0.382, 6.012 | 0.033 | 5 | ok |
| Without 2020 | Negative binomial | 3.159 | 0.743, 5.633 | 0.010 | 5 | warning |
| 2014–2019 | Poisson | 3.673 | 0.212, 7.253 | 0.037 | 4 | ok |
| 2014–2019 | Quasi-Poisson | 3.673 | -0.992, 8.558 | 0.095 | 4 | ok |
| 2014–2019 | Negative binomial | 3.673 | 0.212, 7.253 | 0.037 | 4 | warning |

## Table S7. Pandemic-period influence, age_7_34, IBGE 2018

| Scenario | Series | Years | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|
| Principal | Brazil ASR 7–34 | 8 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013 |
| Principal | Crude 7–34 | 8 | +22.90 | 0.00926 (0.00294, 0.01559) | 0.012 |
| Principal | WHO ASR 7–34 | 8 | +19.83 | 0.00819 (0.00235, 0.01403) | 0.014 |
| Without 2020 | Brazil ASR 7–34 | 7 | +20.99 | 0.00666 (0.00152, 0.01180) | 0.021 |
| Without 2020 | Crude 7–34 | 7 | +22.90 | 0.00710 (0.00188, 0.01232) | 0.017 |
| Without 2020 | WHO ASR 7–34 | 7 | +19.83 | 0.00633 (0.00108, 0.01158) | 0.027 |
| 2014–2019 | Brazil ASR 7–34 | 6 | +24.38 | 0.00767 (-0.00086, 0.01620) | 0.067 |
| 2014–2019 | Crude 7–34 | 6 | +25.45 | 0.00797 (-0.00076, 0.01670) | 0.064 |
| 2014–2019 | WHO ASR 7–34 | 6 | +25.20 | 0.00768 (-0.00087, 0.01622) | 0.067 |

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

## Table S7 continued. Count models, age_7_34, IBGE 2018

| Scenario | Family | APC | 95% CI | p | df | Status |
|---|---|---|---|---|---|---|
| Principal | Poisson | 4.604 | 2.327, 6.931 | <0.001 | 6 | ok |
| Principal | Quasi-Poisson | 4.604 | 1.561, 7.738 | 0.010 | 6 | ok |
| Principal | Negative binomial | 4.604 | 2.327, 6.931 | <0.001 | 6 | warning |
| Without 2020 | Poisson | 3.590 | 1.104, 6.138 | 0.004 | 5 | ok |
| Without 2020 | Quasi-Poisson | 3.590 | 0.975, 6.274 | 0.016 | 5 | ok |
| Without 2020 | Negative binomial | 3.590 | 1.104, 6.138 | 0.004 | 5 | warning |
| 2014–2019 | Poisson | 4.153 | 0.584, 7.849 | 0.022 | 4 | ok |
| 2014–2019 | Quasi-Poisson | 4.153 | -0.220, 8.718 | 0.058 | 4 | ok |
| 2014–2019 | Negative binomial | 4.153 | 0.583, 7.849 | 0.022 | 4 | warning |

## Table S7. Pandemic-period influence, age_5_34, IBGE 2024

| Scenario | Series | Years | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|
| Principal | Brazil ASR 5–34 | 8 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014 |
| Principal | Crude 5–34 | 8 | +21.07 | 0.00841 (0.00249, 0.01433) | 0.013 |
| Principal | WHO ASR 5–34 | 8 | +18.55 | 0.00744 (0.00193, 0.01294) | 0.016 |
| Without 2020 | Brazil ASR 5–34 | 7 | +20.19 | 0.00643 (0.00087, 0.01199) | 0.031 |
| Without 2020 | Crude 5–34 | 7 | +21.07 | 0.00662 (0.00105, 0.01220) | 0.028 |
| Without 2020 | WHO ASR 5–34 | 7 | +18.55 | 0.00596 (0.00036, 0.01155) | 0.041 |
| 2014–2019 | Brazil ASR 5–34 | 6 | +24.27 | 0.00740 (-0.00188, 0.01669) | 0.091 |
| 2014–2019 | Crude 5–34 | 6 | +24.69 | 0.00754 (-0.00180, 0.01687) | 0.088 |
| 2014–2019 | WHO ASR 5–34 | 6 | +24.37 | 0.00720 (-0.00201, 0.01642) | 0.096 |

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

## Table S7 continued. Count models, age_5_34, IBGE 2024

| Scenario | Family | APC | 95% CI | p | df | Status |
|---|---|---|---|---|---|---|
| Principal | Poisson | 4.178 | 1.964, 6.441 | <0.001 | 6 | ok |
| Principal | Quasi-Poisson | 4.178 | 1.303, 7.135 | 0.012 | 6 | ok |
| Principal | Negative binomial | 4.178 | 1.964, 6.441 | <0.001 | 6 | warning |
| Without 2020 | Poisson | 3.339 | 0.919, 5.816 | 0.007 | 5 | ok |
| Without 2020 | Quasi-Poisson | 3.339 | 0.551, 6.203 | 0.027 | 5 | ok |
| Without 2020 | Negative binomial | 3.339 | 0.919, 5.816 | 0.007 | 5 | warning |
| 2014–2019 | Poisson | 3.899 | 0.431, 7.487 | 0.027 | 4 | ok |
| 2014–2019 | Quasi-Poisson | 3.899 | -0.768, 8.786 | 0.082 | 4 | ok |
| 2014–2019 | Negative binomial | 3.899 | 0.431, 7.487 | 0.027 | 4 | warning |

## Table S7. Pandemic-period influence, age_7_34, IBGE 2024

| Scenario | Series | Years | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|
| Principal | Brazil ASR 7–34 | 8 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011 |
| Principal | Crude 7–34 | 8 | +24.37 | 0.00977 (0.00334, 0.01620) | 0.010 |
| Principal | WHO ASR 7–34 | 8 | +22.11 | 0.00889 (0.00290, 0.01487) | 0.011 |
| Without 2020 | Brazil ASR 7–34 | 7 | +23.26 | 0.00732 (0.00202, 0.01262) | 0.016 |
| Without 2020 | Crude 7–34 | 7 | +24.37 | 0.00757 (0.00225, 0.01288) | 0.015 |
| Without 2020 | WHO ASR 7–34 | 7 | +22.11 | 0.00696 (0.00164, 0.01227) | 0.020 |
| 2014–2019 | Brazil ASR 7–34 | 6 | +26.32 | 0.00837 (-0.00043, 0.01716) | 0.057 |
| 2014–2019 | Crude 7–34 | 6 | +26.85 | 0.00853 (-0.00033, 0.01739) | 0.056 |
| 2014–2019 | WHO ASR 7–34 | 6 | +26.85 | 0.00831 (-0.00035, 0.01697) | 0.056 |

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

## Table S7 continued. Count models, age_7_34, IBGE 2024

| Scenario | Family | APC | 95% CI | p | df | Status |
|---|---|---|---|---|---|---|
| Principal | Poisson | 4.791 | 2.511, 7.122 | <0.001 | 6 | ok |
| Principal | Quasi-Poisson | 4.791 | 1.742, 7.931 | 0.008 | 6 | ok |
| Principal | Negative binomial | 4.791 | 2.511, 7.122 | <0.001 | 6 | warning |
| Without 2020 | Poisson | 3.778 | 1.288, 6.329 | 0.003 | 5 | ok |
| Without 2020 | Quasi-Poisson | 3.778 | 1.150, 6.474 | 0.014 | 5 | ok |
| Without 2020 | Negative binomial | 3.778 | 1.288, 6.329 | 0.003 | 5 | warning |
| 2014–2019 | Poisson | 4.391 | 0.814, 8.096 | 0.016 | 4 | ok |
| 2014–2019 | Quasi-Poisson | 4.391 | 0.018, 8.956 | 0.049 | 4 | ok |
| 2014–2019 | Negative binomial | 4.391 | 0.814, 8.096 | 0.016 | 4 | warning |

## Table S8. One-condition sensitivities, ge7, IBGE 2018

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Principal | 18544 | 1.12274 | 1.07718 | -4.06 | 0.00016 (-0.02139, 0.02170) | 0.986 |
| Days/365.25 | 18544 | 1.12274 | 1.07719 | -4.06 | 0.00016 (-0.02139, 0.02170) | 0.986 |
| Original COVID | 18545 | 1.12274 | 1.07772 | -4.01 | 0.00020 (-0.02130, 0.02171) | 0.982 |
| Retain COVID | 18601 | 1.12274 | 1.09149 | -2.78 | 0.00210 (-0.01965, 0.02385) | 0.821 |
| Historical key | 18543 | 1.12274 | 1.07718 | -4.06 | 0.00015 (-0.02142, 0.02172) | 0.987 |

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

## Table S8. One-condition sensitivities, age_5_34, IBGE 2018

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Principal | 1590 | 0.18176 | 0.21453 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018 |

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Days/365.25 | 1590 | 0.18176 | 0.21453 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018 |
| Original COVID | 1591 | 0.18176 | 0.21553 | +18.58 | 0.00760 (0.00195, 0.01324) | 0.017 |
| Retain COVID | 1596 | 0.18176 | 0.21960 | +20.82 | 0.00800 (0.00246, 0.01353) | 0.012 |
| Historical key | 1590 | 0.18176 | 0.21453 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018 |

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

## Table S8. One-condition sensitivities, age_7_34, IBGE 2018

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Principal | 1519 | 0.18146 | 0.21955 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013 |
| Days/365.25 | 1519 | 0.18146 | 0.21955 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013 |
| Original COVID | 1520 | 0.18146 | 0.22060 | +21.57 | 0.00884 (0.00275, 0.01493) | 0.012 |
| Retain COVID | 1525 | 0.18146 | 0.22493 | +23.96 | 0.00926 (0.00331, 0.01522) | 0.009 |
| Historical key | 1519 | 0.18146 | 0.21955 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013 |

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

## Table S8. One-condition sensitivities, ge7, IBGE 2024

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Principal | 18544 | 1.12844 | 1.11123 | -1.53 | 0.00393 (-0.01762, 0.02547) | 0.671 |
| Days/365.25 | 18544 | 1.12844 | 1.11123 | -1.53 | 0.00393 (-0.01762, 0.02547) | 0.671 |
| Original COVID | 18545 | 1.12844 | 1.11178 | -1.48 | 0.00397 (-0.01753, 0.02547) | 0.667 |
| Retain COVID | 18601 | 1.12844 | 1.12583 | -0.23 | 0.00590 (-0.01590, 0.02769) | 0.532 |
| Historical key | 18543 | 1.12844 | 1.11123 | -1.53 | 0.00392 (-0.01765, 0.02549) | 0.672 |

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

## Table S8. One-condition sensitivities, age_5_34, IBGE 2024

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Principal | 1590 | 0.18281 | 0.21972 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014 |
| Days/365.25 | 1590 | 0.18281 | 0.21972 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014 |
| Original COVID | 1591 | 0.18281 | 0.22073 | +20.74 | 0.00827 (0.00246, 0.01409) | 0.013 |
| Retain COVID | 1596 | 0.18281 | 0.22498 | +23.06 | 0.00869 (0.00299, 0.01439) | 0.010 |
| Historical key | 1590 | 0.18281 | 0.21972 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014 |

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

## Table S8. One-condition sensitivities, age_7_34, IBGE 2024

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Principal | 1519 | 0.18258 | 0.22505 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011 |
| Days/365.25 | 1519 | 0.18258 | 0.22505 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011 |

| Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p |
|---|---|---|---|---|---|---|
| Original COVID | 1520 | 0.18258 | 0.22612 | +23.85 | 0.00956 (0.00329, 0.01583) | 0.010 |
| Retain COVID | 1525 | 0.18258 | 0.23063 | +26.32 | 0.01000 (0.00387, 0.01614) | 0.007 |
| Historical key | 1519 | 0.18258 | 0.22505 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011 |

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

## Supplementary Figure S1

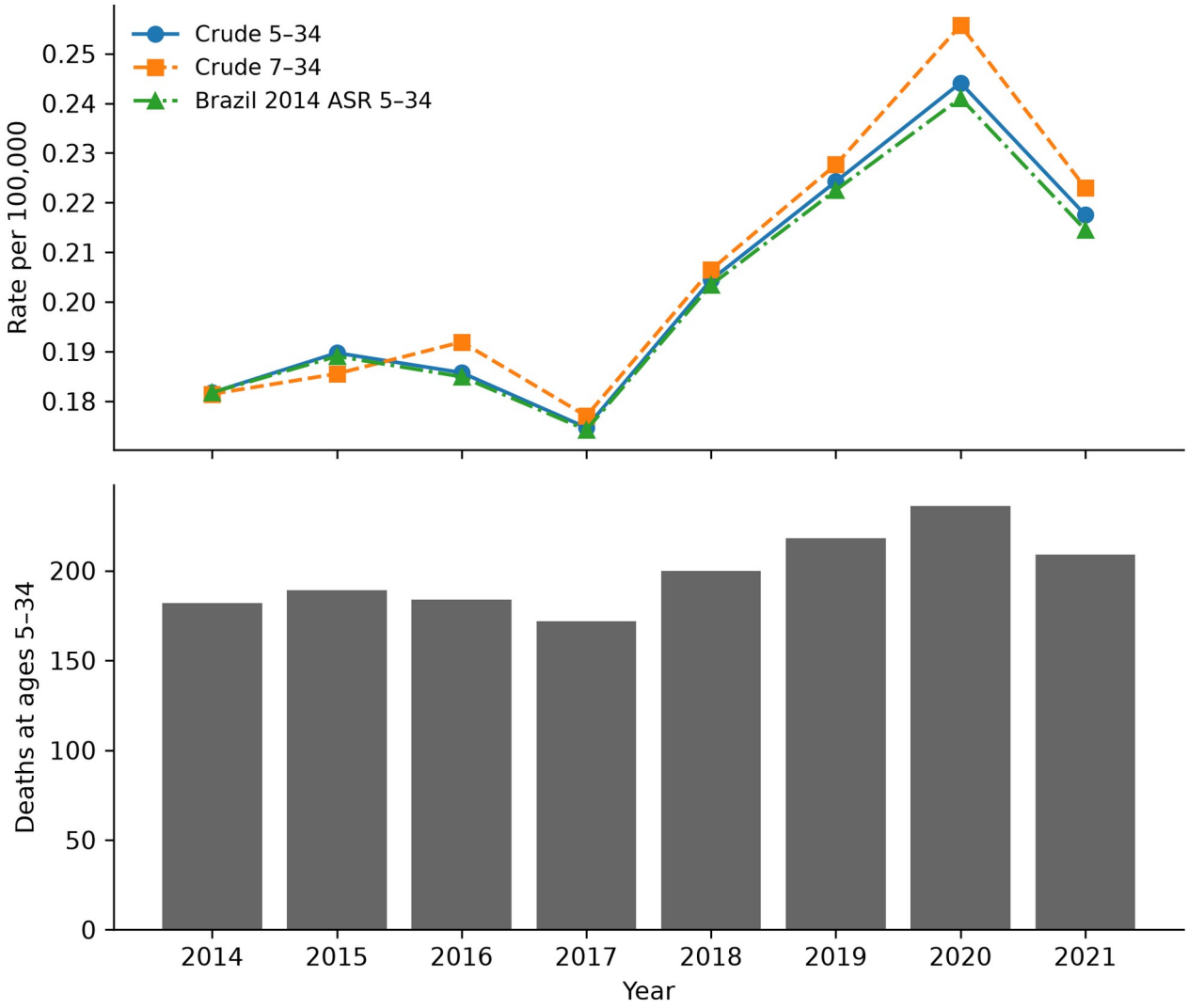



**Figure S1. Annual crude rates for ages 5–34 and 7–34, within-5–34 Brazil-2014-standardised rates, and 5–34 death counts; corrected selection and IBGE 2018.**